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Semantic memory involvement in the default mode network: a functional neuroimaging study using independent component analysis

Miranka Wirth¹, Kay Jann, Thomas Dierks, Andrea Federspiel, Roland Wiest, Helge Horn

Affiliations

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Abstract

The Default Mode Network (DMN) is a higher order functional neural network that displays activation during passive rest and deactivation during many types of cognitive tasks. Accordingly, the DMN is viewed to represent the neural correlate of internally-generated self-referential cognition. This hypothesis implies that the DMN requires the involvement of cognitive processes, like declarative memory. The present study thus examines the spatial and functional convergence of the DMN and the semantic memory system. Using an active block-design functional Magnetic Resonance Imaging (fMRI) paradigm and Independent Component Analysis (ICA), we trace the DMN and fMRI signal changes evoked by semantic, phonological and perceptual decision tasks upon visually-presented words. Our findings show less deactivation during semantic compared to the two non-semantic tasks for the entire DMN unit and within left-hemispheric DMN regions, i.e., the dorsal medial prefrontal cortex, the anterior cingulate cortex, the retrosplenial cortex, the angular gyrus, the middle temporal gyrus and the anterior temporal region, as well as the right cerebellum. These results demonstrate that well-known semantic regions are spatially and functionally involved in the DMN. The present study further supports the hypothesis of the DMN as an internal mentation system that involves declarative memory functions.

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